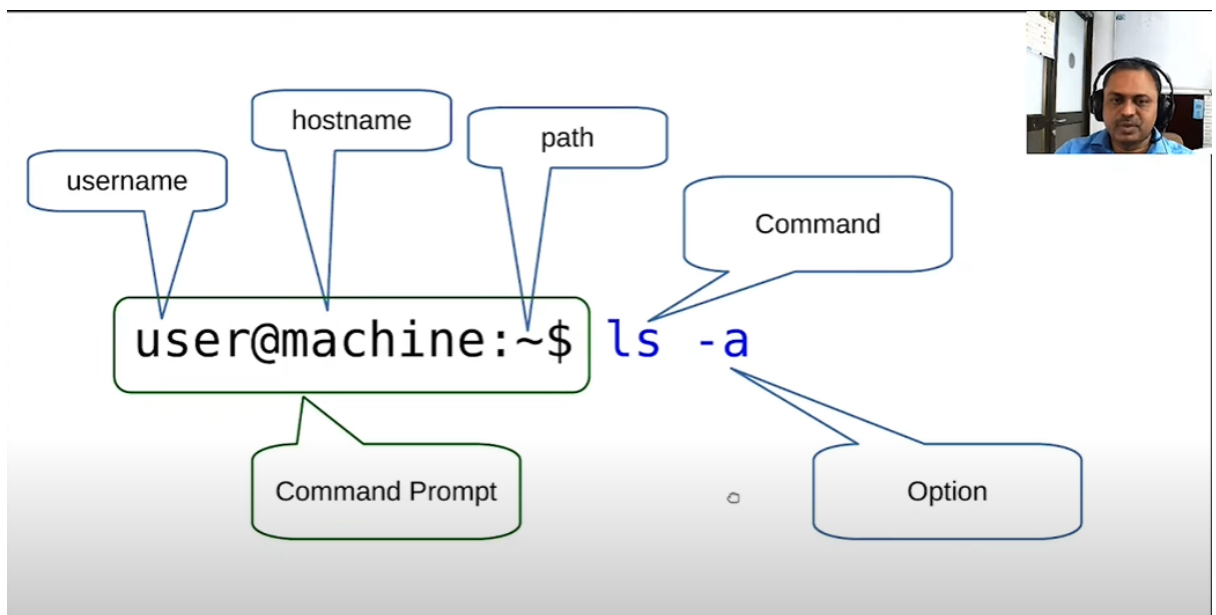


SYSTEM COMMANDS

WEEK 1

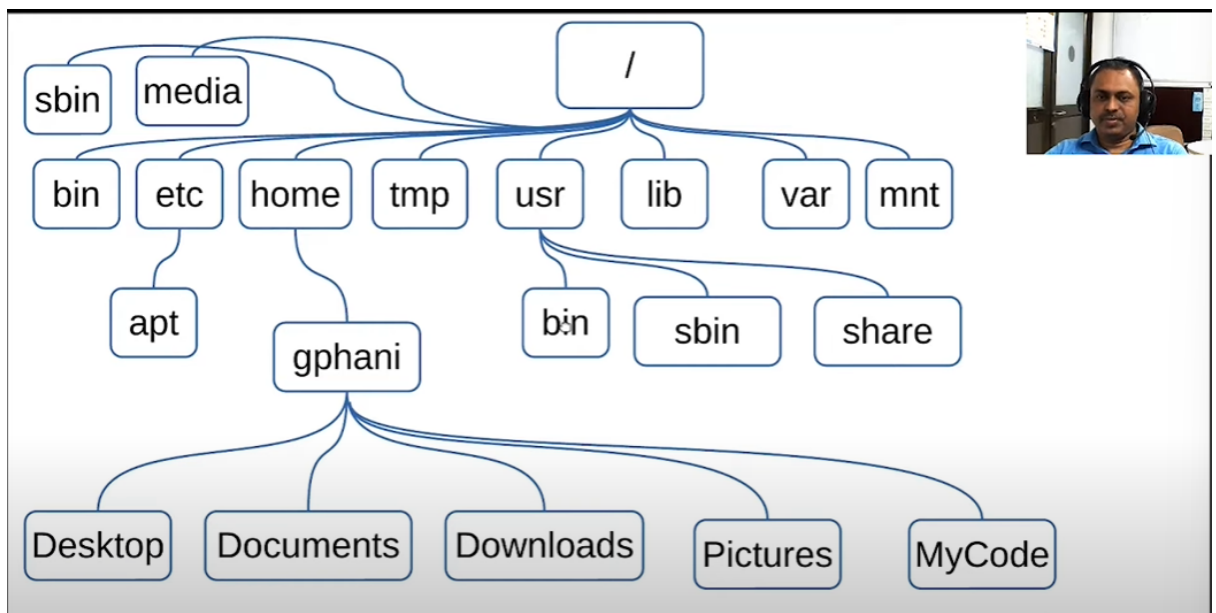
LEC 2: Command Line Environment

- Different Terminal emulators on Linux:
 - Terminal (Default)
 - Konsole
 - Xterm
 - Guake (can be activated and deactivated by F12)
- Commands:
 1. **pwd**: present working directory
 2. **ls**: lists the names of all files/folders in current directory
 3. **ps**: processes that are running presently
 4. **uname**: name of the OS we're currently using
 5. **clear**: clears the screen (Ctrl + L)
 6. **exit**: exits the shell (Ctrl + D)
 7. **cd**: change working directory (can traverse only one level at a time)
- Anatomy:



- Delving into the 'ls' command: (most used are -a, -l)
 - **ls -a**: lists files which have '.' in front of them (i.e., they are usually hidden from users)
 - **ls -l**: files are displayed in a long format
- Help command: 'man' (stands for manual)

- o Pass any command as a parameter with the 'man' command for details regarding its usage. E.g.: ~\$ man ls command gives details about the ls command (Here, 'man' is the command, 'ls' is the argument)
- o Different page sections of man command: (Section can be specified using an option before the argument is specified. E.g.: man 1 ls will display Section 1)
 - Section 1: Executable programs and shell commands
 - Section 2: System calls provided by kernels
 - Section 3: Library calls
 - Section 4: Special files usually found in /dev
 - Section 5: File formats and conventions
 - Section 6: Games
 - Section 7: Misc – macro packages, conventions
 - Section 8: System admin commands
 - Section 9: Kernel routines
- File system of Linux OS:
 - o FHS (Filesystem Hierarchy Standard) 3.0
 - o / (traversing the tree)
 - root directory
 - starting point in the system hierarchy
 - used as a field separator to indicate the path to any directory
 - delimiter for sub-directories
 - o '.' is the current directory
 - o '..' is the parent directory
 - o E.g., of a filesystem hierarchy:



- o Path of traversal can be absolute or relative
- o **cd .** will not perform any changes, whereas **cd ..** will take the pwd to the parent directory. (Goes one level up in the filesystem directory)

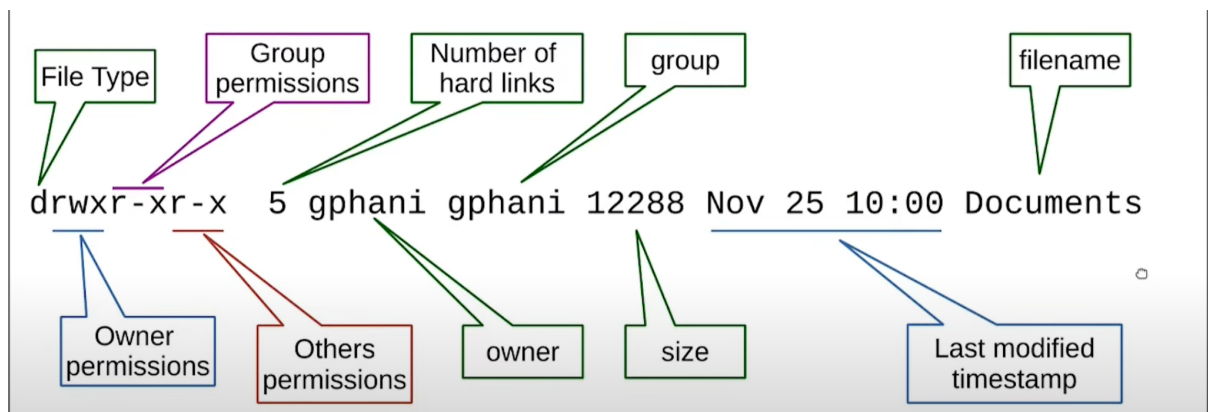
- o cd without any arguments will take the user to the home directory
- ls -l gives the long format of files and it is represented by a set of letters. The 1st letter: **l/d/c/b**
 - 'l' indicates a symbolic link (eg: bin folder of the root directory is a symbolic link to a folder with the same name in the user directory)
 - 'd' indicates that it is a directory by itself in that particular location (eg: boot directory is where the Linux kernel is located, which gets executed when the system starts)
 - 'c' is a character file, which means the user can read character by character from the file.
 - 'b' indicates block devices which are particularly for hard disks. Reads one block at a time (512 bytes/ 1 kB/ 4 kB – depends on the config. In Linux it is 1 kB) [sda1 -sda7 are block devices available as files]
- Organization of folders according to file hierarchy standards:
 - o **/bin**: Essential command binaries/executables
 - o **/boot**: Static files of the boot loader. Runs when the system starts
 - o **/dev**: Device files
 - o **/etc**: Host specific system configuration
 - o **/lib**: Essential shared libraries and kernel modules
 - o **/media**: Mount points for removable devices. Disappears when device is removed
 - o **/mnt**: Mount points
 - o **/opt**: Add on application software packages
 - o **/run**: Data relevant to running processes
 - o **/sbin**: Essential system binaries, normal users aren't required to execute commands from here
 - o **/srv**: Data for services (like FTP/HTTP)
 - o **/tmp**: Temporary files (user has write perms. On reboot, all files here will be deleted)
 - o **/usr**: Secondary hierarchy (shareable across multiple computers)
 - o **/var**: Variable data (such as log files)
- /usr hierarchy:
 - o **/usr/bin**: User commands
 - o **/usr/lib**: Libraries
 - o **/usr/local**: Local hierarchy (for the particular OS)
 - o **/usr/sbin**: Non-vital system binaries
 - o **/usr/share**: Architecture dependent data
 - o **/usr/include**: Header files included by C programs
 - o **/usr/src**: Source code
- /var hierarchy:
 - o **/var/cache**: Application cache data
 - o **/var/lib**: Variable state information

- **/var/local:** Variable data for /usr/local
- **/var/lock:** Lock files
- **/var/log:** Log files and directories
- **/var/run:** Data relevant to running processes
- **/var/tmp:** Temporary files preserved between reboots
- Static/Variable v/s shareable/unshareable directories

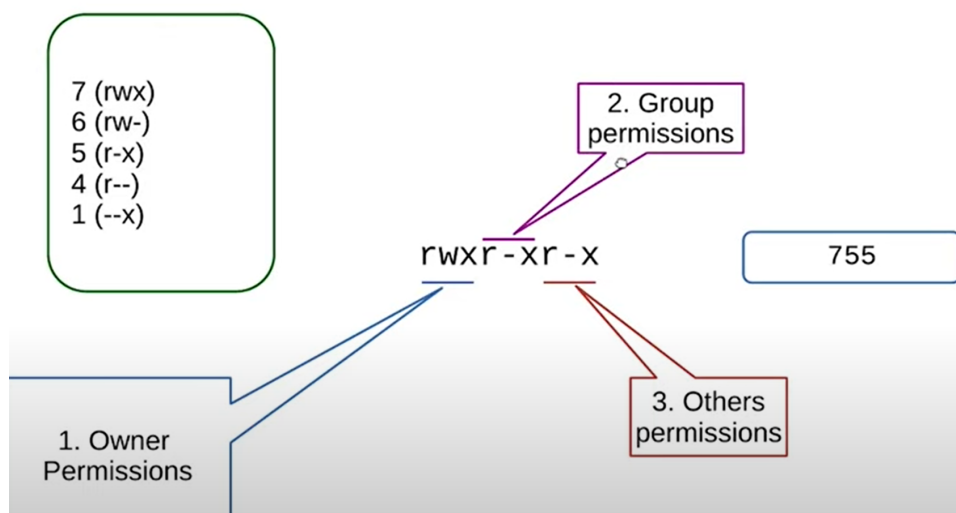
	SHARABLE	UNSHARABLE
STATIC	/usr /opt	/etc /boot
VARIABLE	/var/mail	/var/run /var/lock

LEC 3: Simple Commands in Linux - 1

- Commands:
 - **date:** Date and time
 - **date -R:** gives date and time in the RFC 5322 standard (used for email communications)
 - **cal:** Calendar of current month
 - **cal [-month/month number][-year]:** would display the calendar of the specified month and year.
 - **ncal:** changes orientation of the calendar
 - **free:** Memory statistics
 - **free -h:** To make the numbers human readable
 - **groups:** Groups to which user belongs
 - **file:** What type of a file is it?
 - **cd - :** takes user to the last working directory
 - **cd ~ :** takes user to home directory (other than cd)
 - **groups:** groups that the current user is belonging to
 - Eg: if the user is belonging to a group called “sudo” it means the user is also a super user (system admin)
 - There is a group name with the name of the user so as to provide privacy to certain files
 - **mkdir:** Create a new folder/directory
- Typical output of ls -l:



- File Types
 - **-** : Regular file (script/text/exe files)
 - **d**: Directory (directory under which there are other files/directories)
 - **l**: Symbolic link
 - **c**: Character file (typically the terminal)
 - **b**: Block file (typically the hard disk)
 - **s**: Socket file (process that is happening behind, 2-way, can listen to the communication - not a static file)
 - **p**: Named pipe (process behind to establish one way communication - only read, can't send)
- Hard links
 - **inode**: `ls -li <name>`
 - An entry in the filesystem table about the location in storage media
 - Unique for unique files
- Permission string:



- Changing permissions:
 - Using **chmod**:

- **Syntax:** `chmod <if group - g, user - u, others - o > <remove perms- ; add perms +> <r,w,x> <name of directory/file>`
 - eg: `chmod g -w level1` will remove the write permission for the group for the directory level1
 - If the user want perms only for owner:-
 - `chmod 700 <name>`
- Files and folders:
 - Creating files using **touch**: command used to change the timestamp of the last modified info about the file. If the file doesn't exist, it creates a new file.
 - Copy files into another file using **cp**: `cp <file to be copied> <where it is to be copied>`
 - Move the file one level up using **mv**: `mv <file name> <./..>`
 - `..` means the parent directory
 - Can also be used to rename files: `mv <old filename> <new filename>`
 - If the user wants to name the file using more than 1 word then use quotation marks e.g.: `mv file1 "file one"`
 - For such files, to change the perms, give the file name within quotations. eg: `chmod 700 "file one"`
 - Remove files using **rm**: Removes files. This command by default deletes the file without asking for confirmation. But using the **alias** command we can change the default setting of `rm` command to `"rm -i"` which is done by:
 - `alias rm="rm -i"`
- Hardlinks:
 - Hardlinks are the numbers at the end of the long representation of files
 - can be found using inode number
 - `ls -li` gives the inode number of the directory
 - `ls -lia` gives the inode numbers including the `.` and `..`
 - When this is executed, the inode number of the directory is the same as the inode number of the `.` directory
 - The number of hardlinks increase by 1 as we keep creating sub directories
 - Hardlink gives the number of different entries of the file is there in the file system
 - If the first digit/letter of the long representation is a `'-'`, then number of hardlinks will be 1
- Alias:
 - `cd ~` for going to home directory
 - `cd -` for going back to the previous directory
 - `whoami` for printing current user
- User does not have write permissions to create or delete files from root directory

- If a file in the root directory has read perms, the user can read it using **less** command.
 - `less <filename>`
- The **file** command tells the type of file
 - Syntax: `file <filename>`

<code>chmod</code>	Change permissions of a file
<code>touch</code>	Change modified timestamp of a file
<code>cp</code>	Create a copy of a file
<code>mv</code>	Rename or move a file
<code>mkdir</code>	Create a directory
<code>rm</code>	Remove a file